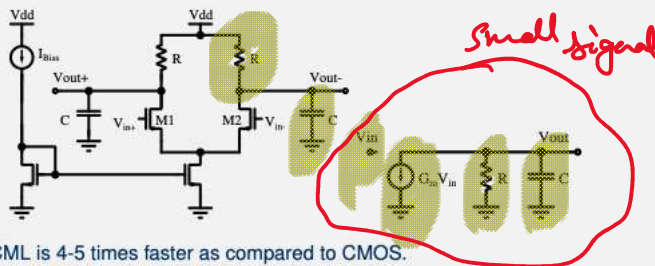


II CML circuits design

Preferable circuits:

Differential amplifier

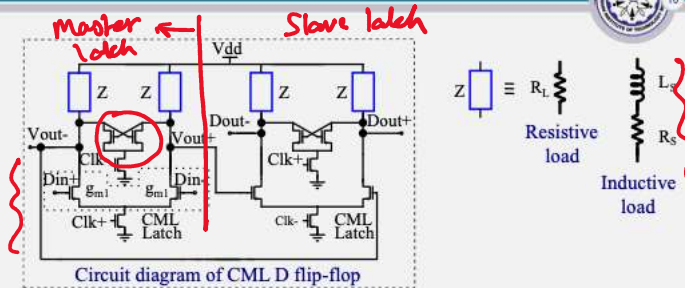


CML is 4-5 times faster as compared to CMOS.

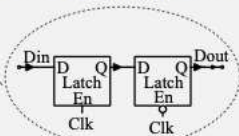
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Broadband circuit design techniques

Circuit diagram of D flip-flop



Circuit diagram of CML D flip-flop



Block diagram of CML D flip-flop

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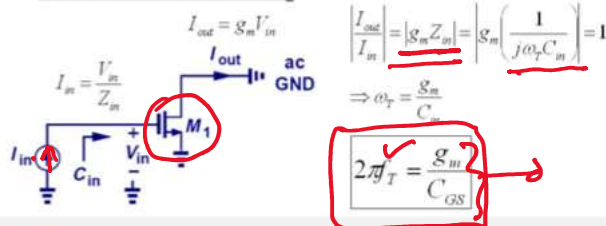
Broadband circuit design techniques

Transit frequency f_T

Transit Frequency

- The "transit" or "cut-off" frequency, f_T , is a measure of the intrinsic speed of a transistor, and is defined as the frequency where the current gain falls to 1.

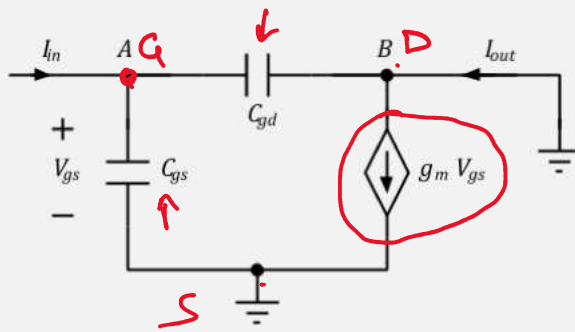
Conceptual set-up to measure f_T



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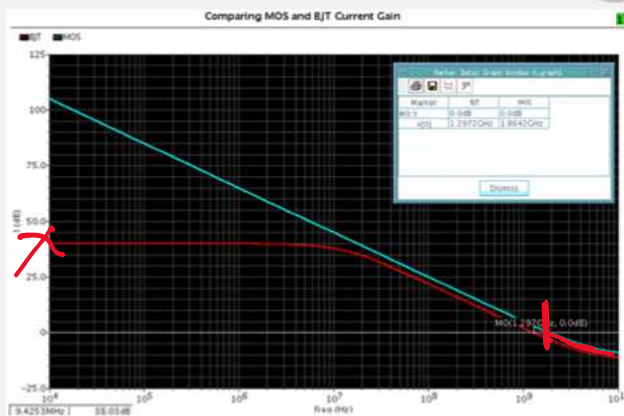
Transit frequency f_T



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Transit frequency f_T

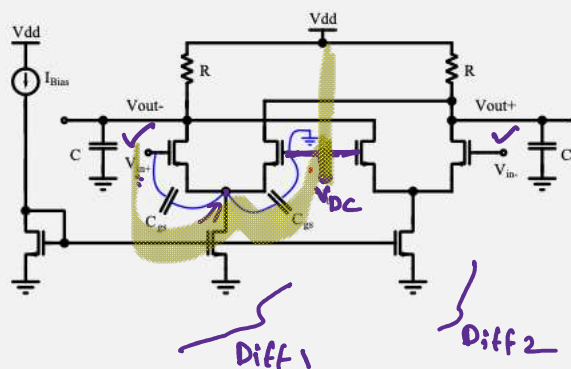


60-litg }
180nm } Technology node
65nm }
↓
1204H₃=fT

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Broadband circuit design techniques

f_T doubler

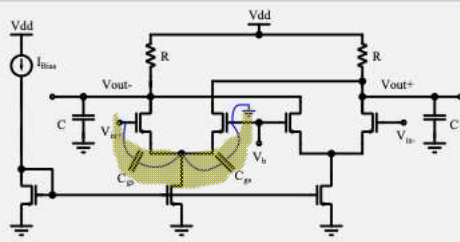


A hand-drawn circuit diagram on a blue grid background. It shows a voltage divider circuit. A voltage source V_{in} is connected in series with two resistors. The output voltage V_{out} is taken across the second resistor. The output is labeled "Virtual ground".

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Broadband circuit design techniques

f_T doubler



- f_T doubler technique for reducing gate to source capacitance to half.

$$f_{T_{new}} = \frac{g_m}{2\pi C_{gs}} = \frac{2g_m}{2\pi C_{gs}} = 2f_T$$

doubles the speed

- Power consumption doubles.

[1] Heydari et al. TVLSI-2004. [6] Mohanavelu et al. ISCAS-2004. [7] Ray et al. JSSC-2016. [8] Razavi GaAs ICs-2001. [9] Razavi JSSC-2002.

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Broadband circuit design techniques

Inductive peaking

Inductive peaking

(a) Using Passive components

- 1 Shunt peaking.
- 2 Series peaking.
- 3 Shunt-series peaking.
- 4 Shunt-double series peaking (T-coil)

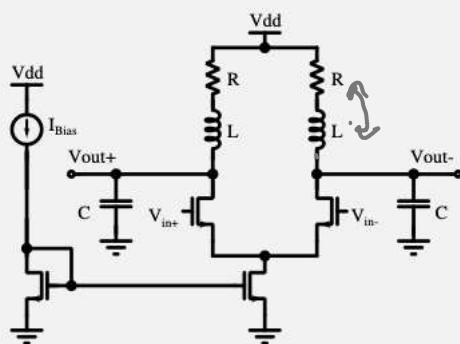
(b) Active inductors.

[1] Heydari et al. TVLSI-2004. [4] Galal et al. USSC-2003. [10] Feucht 1990. [11] Razavi et al. JSSC-2003. [3] Wohlmuth et al. ICSD-2004. [12] Dokson et al. CSC-2004. [13] Dokson et al. JSSC-2007. [14] Staric et al. book-2007. [15] Kim et al. CICC-2003. [8] Razavi et al. GaAs ICs-2001. [6] Razavi et al. JSSC-2002. [16] Singh et al. JSSC-2005. [17] Buren et al. ESSC-2006. [18] Chelvatzi et al. JSSC-2007.

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Broadband circuit design techniques

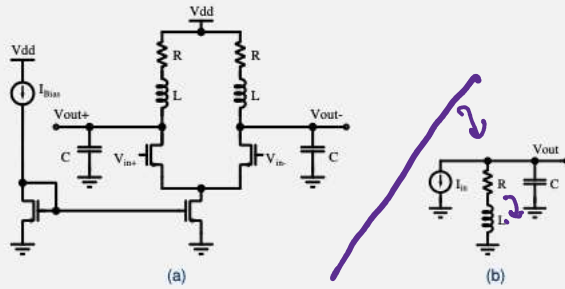
Shunt peaking



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Broadband circuit design techniques

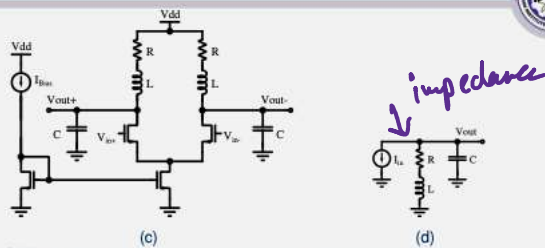
Shunt peaking



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Broadband circuit design techniques

Shunt peaking



- (a) A CML buffer with inductor in series with resistive load;
(b) Its equivalent circuit.

The output impedance is given by

$$\frac{V_{out}}{I_{in}} = \frac{s(L/R) + 1}{s^2LC + sRC + 1} \quad \left\{ \frac{A_{vo}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right\}$$

Here, $\zeta = \frac{RL\sqrt{C}}{2}$, and $\omega_n = \frac{1}{\sqrt{LC}}$.

[1]Heydari et al TVLSI-2004, [4]Galal et al USSC-2003, [10]Feucht 1990, [11]Razavi et al JSSC-2003, [3]Wohlmuth et al ICSD-2004, [12]Dickson et al CSC-2004, [13]Dickson et al JSSCC-2007, [14]Stancic et al book-2007, [15]Kim et al CICC-2003, [8]Razavi et al GaAs ICS-2001, [6]Razavi et al JSSC-2002, [16]Singh et al JSSC-2005, [17]Buren et al ESSCC-2006, [18]Chalvatzi et al JSSCC-2007.

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Broadband circuit design techniques

Shunt peaking

continue

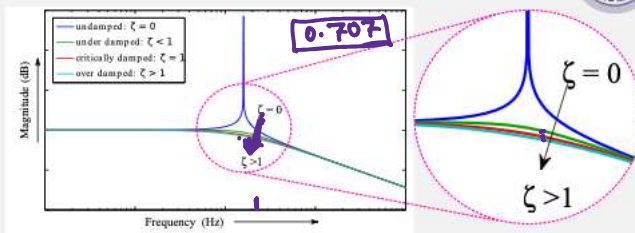


Figure: frequency domain waveform for various ζ for second order equations.

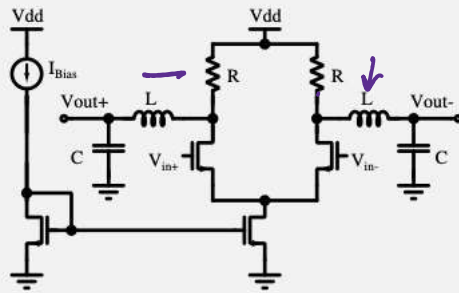
- ▶ Inductive peaking $\Rightarrow$ parallel resonance,
- ▶ $\zeta \Rightarrow 0.5$ to 0.9 for stable operation.
- ▶ Shunt peaking $\Rightarrow 1.6$ times improvement in bandwidth as compared to without inductor load.

[1]Heydari et al TVLSI-2004, [4]Galal et al USSC-2003, [10]Feucht 1990, [11]Razavi et al JSSC-2003, [3]Wohlmuth et al ICSD-2004, [12]Dickson et al CSC-2004, [13]Dickson et al JSSCC-2007, [14]Stancic et al book-2007, [15]Kim et al CICC-2003, [8]Razavi et al GaAs ICS-2001, [6]Razavi et al JSSC-2002, [16]Singh et al JSSC-2005, [17]Buren et al ESSCC-2006, [18]Chalvatzi et al JSSCC-2007.

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Broadband circuit design techniques

Series peaking



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Series peaking

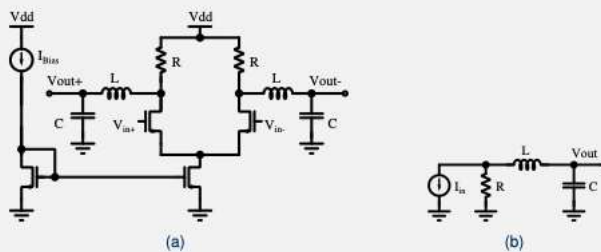


Figure: (a) A CML buffer with inductor in series with resistive load ; (b) Its equivalent circuit.

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Broadband circuit design techniques

Series peaking

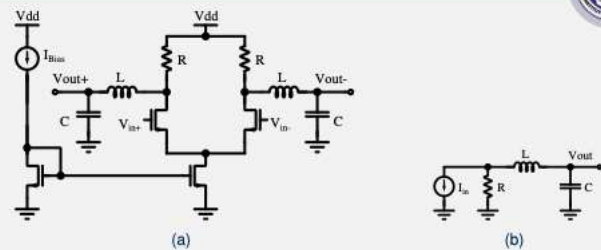


Figure: (a) A CML buffer with inductor in series with resistive load ; (b) Its equivalent circuit.

The output impedance is given by

$$\frac{V_{out}}{I_{in}} = \frac{R}{s^2LC + sRC + 1}$$

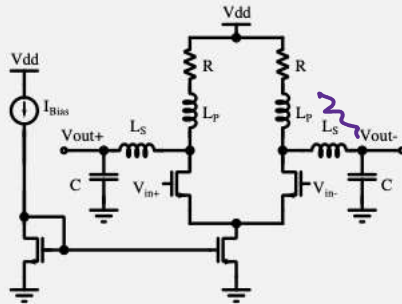
Here, $\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$, and $\omega_n = \frac{1}{\sqrt{LC}}$.

[1] Heydari et al. TVLSI-2004, [4] Galal et al. JSSC-2003, [10] Feucht 1990, [11] Razavi et al. JSSC-2003, [3] Wohlmuth et al. ICSD-2004, [13] Dokken et al. CICC-2004, [13] Dokken et al. JSSCC-2007, [14] Stancu et al. book-2007, [15] Kim et al. CICC-2003, [8] Razavi et al. GaAs ICS-2001, [8] Razavi et al. JSSC-2002, [16] Singh et al. JSSC-2005, [17] Buren et al. ESSCC-2006, [18] Chahvatzi et al. JSSCC-2007.

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Broadband circuit design techniques

Shunt-series peaking



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Shunt-series peaking

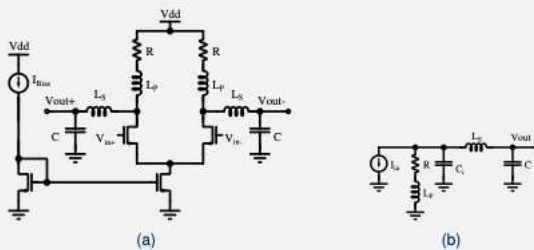


Figure: (a) A CML buffer with inductor in series with resistive load ; (b) Its equivalent circuit.

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Broadband circuit design techniques

Shunt-series peaking

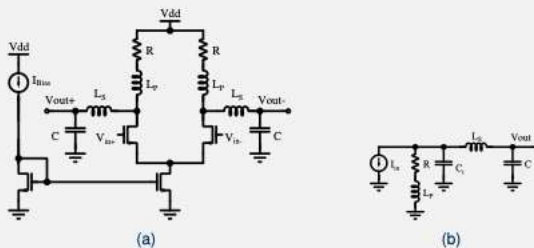


Figure: (a) A CML buffer with inductor in series with resistive load ; (b) Its equivalent circuit.

- ▶ A combination of shunt and series peaking $\Rightarrow$ 2X times bandwidth improvement.

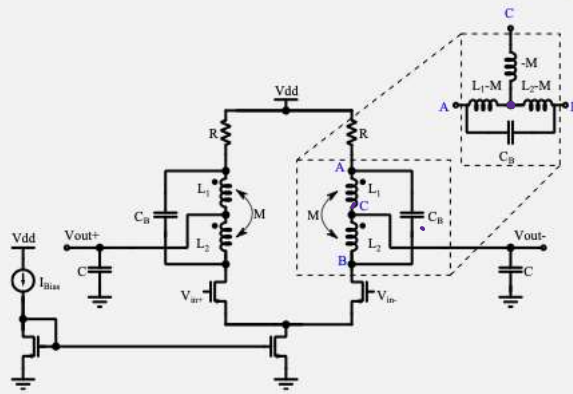
[1] Heydari et al. TVLSI-2004, [4] Galal et al. USSC-2003, [10] Feucht 1990, [11] Razavi et al. JSSC-2003, [3] Wohlmuth et al. ICSD-2004, [13] Dokson et al. CICC-2004, [13] Dickson et al. JSSCC-2007, [14] Staric et al. book-2007, [15] Kim et al. CICC-2003, [8] Razavi et al. GaAs ICS-2001, [8] Razavi et al. JSSC-2002, [16] Singh et al. JSSC-2005, [17] Buren et al. ESSCC-2006, [18] Chahvatzi et al. JSSCC-2007.

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Broadband circuit design techniques

15 db/s
25 db/s-30 db/s

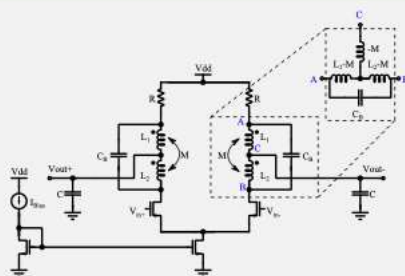
T-coil (Shunt and double series)



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T-coil (Shunt and double series)



$$\frac{V_{out}}{I_{in}} = \frac{R}{\frac{1}{4} \left(\frac{M-1}{M+1} \right) R^2 C^2 s^2 + \frac{1}{2} RCs + 1}$$

$$C_B = \frac{1}{4} \left(\frac{M-1}{M+1} \right) C$$

$$L_1 = L_2 = \frac{R^2 C}{2(M+1)}$$

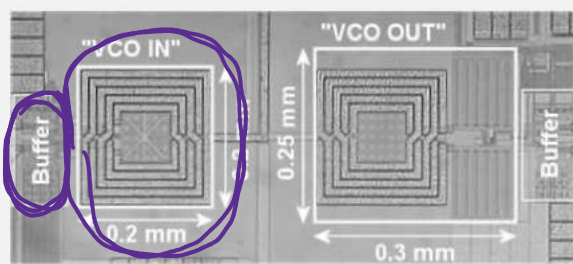
- ▶ For $M=1 \Rightarrow$ single pole system $\Rightarrow$ 1.4X times bandwidth improvement. 3
- ▶ For $M > 1 \Rightarrow$ bandwidth enhancement upto 2.2X. 3

[5] Kim et al. ASSCC-2006. [19] Cheema et al. RFIC-2006. [4] Galal et al. ISSCC-2003. [20] Galal et al. JSSC-2003. [21] Sawoj et al. JSSC-2008.

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Broadband circuit design techniques

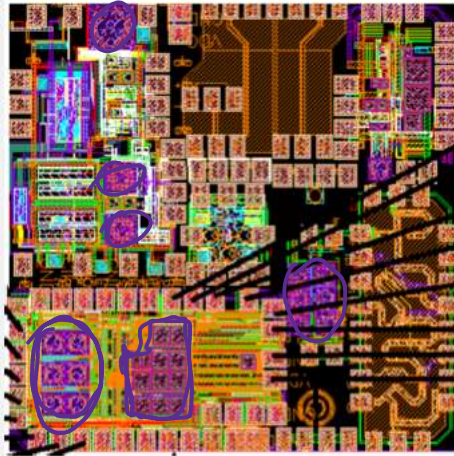
On-chip inductor



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Broadband circuit design techniques

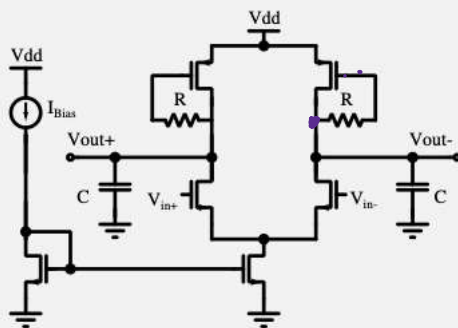
On-chip inductor in a die



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Broadband circuit design techniques

Active inductor



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Broadband circuit design techniques

Active inductor

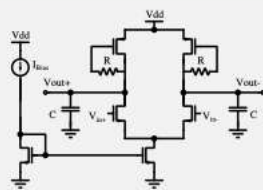


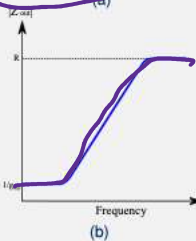
Figure: A CML buffer with active inductor load:-

$$Z_{out} = \frac{RC_{gs}s + 1}{g_{mp} + C_{gs}s}$$

At low frequency $|Z_{out}(s=0)| = 1/g_{mp}$ and at high frequency $|Z_{out}(s=\infty)| = R$. If $R \gg 1/g_{mp}$

$R_A = 1/g_{mp}$ in series with parallel combination of a resistor $R_B = R - 1/g_{mp}$ and inductor $L = \frac{C_{gs}}{g_{mp}}(R - \frac{1}{g_{mp}})$

The quality factor (Q) of inductor $Q \uparrow = R_B/L = \frac{g_{mp}}{C_{gs}}$



[1]Heydari et al TVLSI-2004, [4]Galal et al USSC-2003.

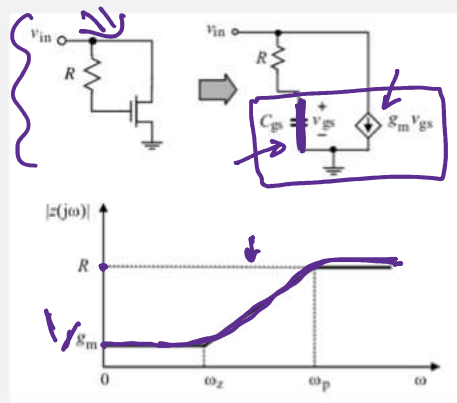
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Broadband circuit design techniques

Active inductor



33



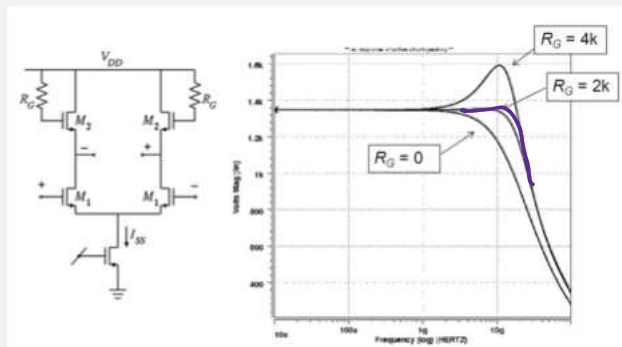
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Broadband circuit design techniques

Active inductor



34



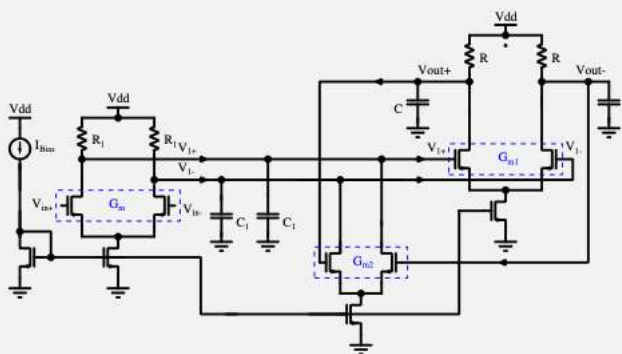
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Broadband circuit design techniques

Active feedback



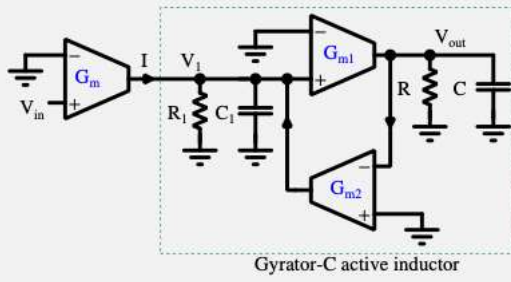
35



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Broadband circuit design techniques

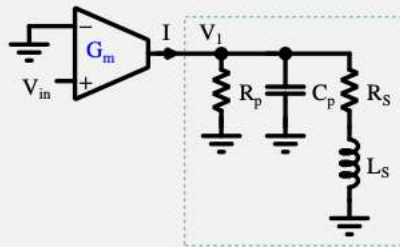
Active feedback



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Broadband circuit design techniques

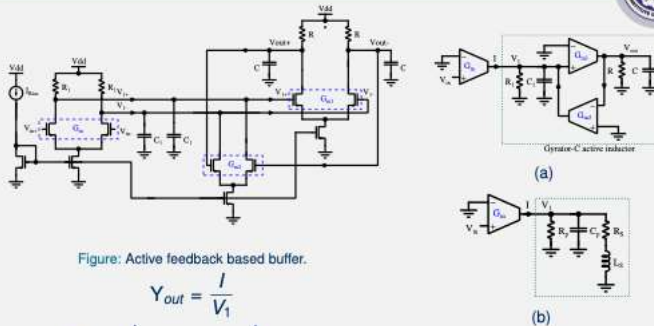
Active feedback



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Broadband circuit design techniques

Active feedback



$$Y_{out} = \frac{I}{V_1}$$

$$= sC_1 + \frac{1}{R_1} + \frac{1}{s\left(\frac{C}{G_{m1}G_{m2}}\right) + \frac{1}{RG_{m1}G_{m2}}}$$

$$R_p = R_1; C_p = C_1; R_s = \frac{1}{RG_{m1}G_{m2}}; L_s = \frac{C}{G_{m1}G_{m2}}$$

[22]Williams et al. JLT-1986, [4]Gatal et al. JSSC-2003, [20]Gatal et al. ISSCC-2003, [22]Samadi et al. ISCAS-2004, [23]Li et al. TCASI-2009, [24]Chen et al. SPO-2010, [25]Baker et al. EL-1996.

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Broadband circuit design techniques

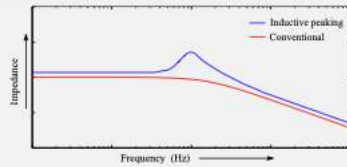


Figure: Variation of impedance with frequency.

► Inductive peaking $\Rightarrow$ parallel resonance.

The resonant frequency

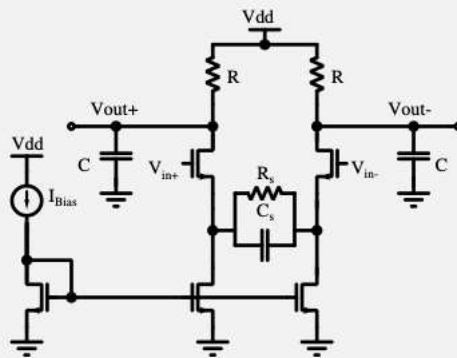
$$\omega_0 = \frac{1}{LC_p} = \sqrt{\frac{G_{m1}G_{m2}}{C_1C_2}}$$

$$A_{v0} = \frac{G_m G_{m1} R_1 R}{1 + G_{m1} G_{m2} R_1 R}; \zeta = \frac{1}{2} \frac{R_1 C_1 + R C_2}{\sqrt{R_1 R C_1 C_2 (1 + G_{m2} G_{m1} R_1 R)}}; \omega = \sqrt{\frac{1 + G_{m2} G_{m1} R_1 R}{R_1 R C_1 C_2}}$$

► $\zeta \Rightarrow 0.707$ for maximally-flat Butterworth response.

[22]Williams et al. JLT-1986. [4]Gatal et al. JSSC-2003. [20]Gatal et al. ISSCC-2003. [22]Samadi et al. ISCAS-2004. [23]Lu et al. TCASI-2009. [24]Chen et al. SPO-2010. [25]Baker et al. EL-1998.

Capacitive degeneration



Capacitive degeneration

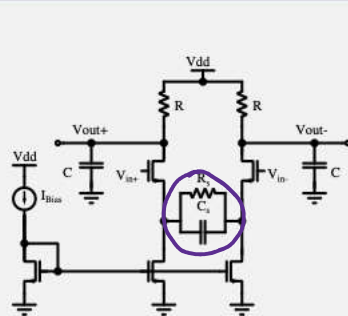
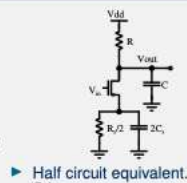
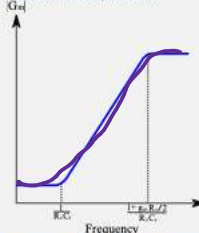


Figure: Capacitive degenerated CML buffer.

$$G_m = \frac{g_m(R_s C_s + 1)}{1 + R_s C_s + g_m R_s / 2}$$



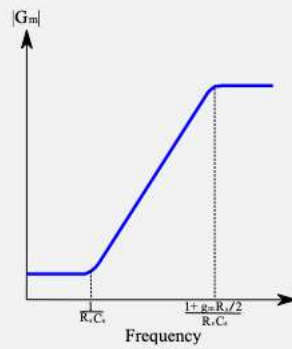
► Half circuit equivalent.



► Variation of G_m with frequency.

[1]Heydari et al. TVLSI-2004. [20]Gatal et al. ISSCC-2003. [4]Gatal et al. USSC-2003. [11]Razavi et al. 2003.

Capacitive degeneration

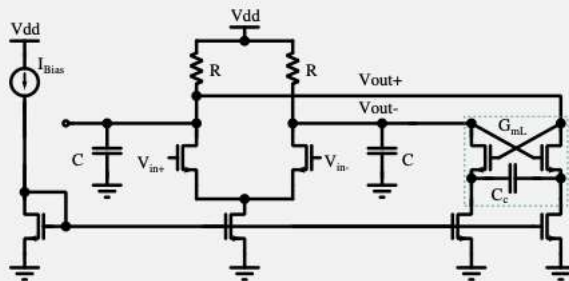


$$G_m = \frac{g_m(R_s C_s s + 1)}{1 + R_s C_s + g_m R_s / 2}$$

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Broadband circuit design techniques

Negative impedance converter



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Broadband circuit design techniques

Negative impedance converter

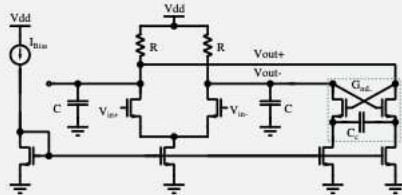


Figure: Buffer with negative impedance converter.

The impedance seen looking into the drains is expressed as

$$Z_{NI} = -\frac{2}{G_{mL}} - \frac{1}{sC_c}$$

Here, G_{mL} = the transconductance of back to back transistors and C_c is the degenerating capacitor.

- it brings negative capacitance ► capacitive output load ↓
- operating frequency ↑.

[1]Heydari et al TVLSI-2004. [4]Galal et al USSC-2003.

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Broadband circuit design techniques

Negative Miller capacitance



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